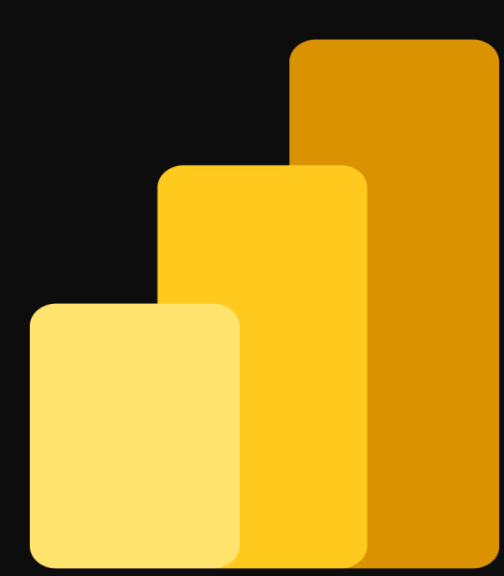


Case Study: Retail Sales Insights Dashboard





Project Context and Objective

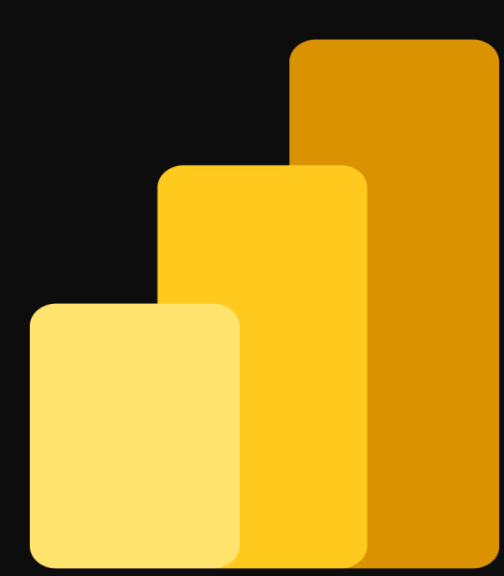
Imagine you've just joined a retail company as a **Data Analyst**. On your first day, your manager gives you your first assignment: "We need a **Sales Insights Dashboard** — something clear, automated, and interactive. It should help us track performance across products, regions, and sales reps. You'll have access to all the raw data — just make sense of it and make it easy for everyone to use." This is both exciting and challenging — a chance to build something from scratch that could directly support decision-making at the company.





Step 1: Understand the Requirements

- Define the goal: build a **one-page Power BI dashboard** to display sales insights across multiple dimensions — product, geography, and time.
- Identify and gather the required data sources:
 - Sales data (stored by year in a folder)
 - Categories (Excel file)
 - SubCategories (Excel file)
 - Sales Representatives (Excel file)
 - Product data (CSV or database)
 - Geography data (Excel file)
- Ensure the system design is **resilient**:
 - Removing a yearly sales file should not cause refresh errors.
 - Adding a new sales file should automatically include it in the dataset upon refresh.



Step 2: Data Preparation

•Task 11 - Automate Sales Data Loading

- Use **Power Query** to create a **dynamic loading mechanism** that imports all yearly sales files from the sales folder.
- Combine all files into a single **Sales Fact Table**.
- Ensure the model automatically updates when new files are added, without requiring manual intervention.



Step 3: Data Cleaning and Transformation

•Task 1.1 – Automate Sales Data Loading

•Task 2.1 – Split and Format Fields

- Split the **Location** field in the Sales table into two separate columns: *City* and *Country*.
- Assign appropriate **data types** for use in Power BI maps.
- Standardize the **Date field** to support time intelligence calculations in DAX.

•Task 2.2 – Create GeoKey

- Create a unique **GeoKey** column in both the **Sales** and **Geography** tables.
- Use this key to establish relationships between sales transactions and geographic locations.

•Task 2.3 – Clean ID Columns

- Identify columns in the **SalesRep** and **SubCategory** tables that contain IDs with the prefix “ID - ”.
- Build a reusable **Power Query function** that removes this prefix.
- Apply the function to both tables to clean and standardize the ID columns.

•Task 2.4 – Build the Data Model

- Design a **star schema** model:
 - Fact Table:** Sales
 - Dimension Tables:** Product, Category, SubCategory, SalesRep, Geography, Calendar
- Validate relationships between tables to ensure accurate aggregation and reporting.



Step 4: Create Business Measures with DAX

- Define DAX measures to calculate key performance metrics:
- **Total Revenue** = Units × Product Retail Price
- **Total Cost** = Units × Product Standard Cost
- **Gross Profit** = Total Revenue – Total Cost
- **Gross Profit MoM Growth %** = Month-over-Month change in Gross Profit
- **Average Sales per Day** = Total Revenue ÷ Number of Active Sales Days
- **QoQ Growth** = Quarter-over-Quarter growth (for QBR reporting)
- Verify all measures for accuracy and consistency using sample data.



Step 5: Design the Dashboard

- Use Power BI visuals to create an **interactive one-page dashboard**.
- Include the following elements:
 - **KPI Cards**: Total Revenue, Total Cost, Gross Profit
 - **Line Charts**: Month-over-Month (MoM) and Quarter-over-Quarter (QoQ) growth
 - **Maps**: Sales distribution by Country and City
 - **Bar Charts**: Top Products and Sales Representatives
 - **Filters/Slicers**: Year, Country, Category
- Ensure months are sorted chronologically from **January to December**.
- Design the layout for clarity, readability, and executive-level presentation.

Step 6: Deploy and Share the Report

- **Publish the Power BI report following these deployment steps:**
- Create a **Power BI Workspace** and assign appropriate user permissions.
- **Publish** the report to the Power BI Service.
- Configure a **Gateway** to enable on-premises data refresh.
- Set up **Scheduled Refreshes** for automated updates.
- Add **Roles, Subscriptions, and Alerts** for stakeholders.
- Package the report into a **Power BI App** and share it across departments.

Step 7: Evaluate Results

- **Verify that the dashboard:**

- Updates automatically when new yearly data is added.
- Provides accurate and real-time performance insights.
- Enables managers to analyze sales trends by product, region, and representative.
- Confirm that the dashboard supports **data-driven decisions** during quarterly business reviews (QBRs).
- Document feedback from management and optimize visuals or metrics if needed.